

21.10.15.

Topic: First Order Computation in Φ^4 Theory

$$S = \Omega^{-1} \Omega_+ = \lim_{\substack{t_1 \rightarrow -\infty \\ t_2 \rightarrow \infty}} \mathcal{U}_0(-t_2) \mathcal{U}(t_2 - t_1) \mathcal{U}_0(t_1)$$

$$H = \underbrace{H_0}_{\text{interaction}} + g H_I$$

hamiltonian

Dyson Series

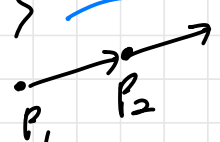
$$S = 1 + \sum_{n=1}^{\infty} \frac{(-ig)^n}{n!} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{T} \underbrace{H(\theta_1) H(\theta_2) \dots}_{d\theta_1 \dots d\theta_n}$$

$$H_I = V, \quad H_0 = -\frac{1}{2m} \Delta$$

$$H = \int \mathcal{H}$$

$$S = 1 + (-ig) \int_{-\infty}^{\infty} e^{i\int H_0} V e^{-i\int H_0} dt + \mathcal{O}(g^2)$$

$$S = 1 + \sum_{n=0}^{\infty} \frac{(-ig)^n}{n!} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \dots \mathcal{T} \underbrace{\mathcal{H}(x_1) \mathcal{H}(x_2) \dots}_{d x_1 \dots d x_n}$$

$\langle P_2 | S | P_1 \rangle$  probability of P_1 becomes P_2 .

$$\langle \vec{P}_1, \vec{P}_2, \dots, \vec{P}_k | S | \vec{Q}_1, \dots, \vec{Q}_j \rangle$$

$$\langle \vec{P}_3, \vec{P}_4 | S | \vec{P}_2, \vec{P}_1 \rangle$$

Notation $P = (\underbrace{u_P}_{\text{time}}, \vec{p}) \in \mathbb{R}^{1,3}$ spacetime.

$$\sqrt{m^2 + |\vec{p}|^2}$$

$$|\vec{p}_1\rangle = \widehat{e^{i\vec{p}_1 \cdot \vec{x}}}$$

$$\langle \vec{p}_2 | \vec{p}_1 \rangle = (2\pi)^3 \delta^{(3)}(\vec{p}_2 - \vec{p}_1)$$

$$\text{Def } |\vec{p}_j, \dots, \vec{p}_1\rangle := a^\dagger(\vec{p}_j) \dots a^\dagger(\vec{p}_1) |0\rangle$$

$\mathcal{H}_{\text{sym}}^{\otimes 2}$

$1 \in \mathbb{C}$

$\uparrow$
 $\mathcal{H}_{\text{sym}}^{\otimes 1}$

$$\langle \vec{q}_1, \dots, \vec{q}_j | := \langle 0 | a(\vec{q}_1) \dots a(\vec{q}_j)$$

$$\cdot \langle \vec{p}_3, \vec{p}_4 | \vec{p}_2, \vec{p}_1 \rangle = \langle 0 | a(\vec{p}_3) a(\vec{p}_4) a^\dagger(\vec{p}_2) a^\dagger(\vec{p}_1) | 0 \rangle$$

$\mathcal{P}_4$ Theory

$$\mathcal{H} = \frac{1}{4!} : \phi(x)^4 :$$

$$\langle \vec{p}_3, \vec{p}_4 : \phi^4 : \vec{p}_2, \vec{p}_1 \rangle$$

$$= \langle 0 | a(\vec{p}_3) a(\vec{p}_4) : \phi^4 : a(\vec{p}_2) a(\vec{p}_1) | 0 \rangle$$

normal ordering term

I. Wick Theorem

$$C := \{ A(\xi) + A^\dagger(\eta) : \xi, \eta \in \mathcal{H} = L^2(X_m, d\lambda_m) \}$$

Def A pairing ℓ of $\overset{\text{even}}{k}$ to is a set of the form

$$\left\{ \begin{aligned} & \{ (\ell_1, \ell'_1), \dots, (\ell_{k/2}, \ell'_{k/2}) \} \text{ s.t. } \ell_j < \ell'_j, \\ & \{ \ell_1, \dots, \ell_k \} = \{ 1, 2, \dots, k \} \end{aligned} \right.$$

ex) $k=4$

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \quad \{ (1 \ 2), (3 \ 4) \}$$

$\underbrace{\hspace{1.5cm}} \quad \underbrace{\hspace{1.5cm}}$

$$\{ (1 \ 3), (2 \ 4) \}$$

$\underbrace{\hspace{2.5cm}} \quad \underbrace{\hspace{2.5cm}}$

$$\{ (1 \ 4), (2 \ 3) \}$$

$\underbrace{\hspace{3.5cm}} \quad \underbrace{\hspace{1.5cm}}$

~~But~~ # of pairings of $k = (k-1)!!$
 $= (k-1)(k-3) \dots 5 \dots 3 \cdot 1.$

<pf>
$$\frac{\binom{k}{2} \binom{k-2}{2} \dots \binom{2}{2}}{\underbrace{\left(\binom{k}{2} \right)!}_{\text{순서 무시}}} \quad \square$$

Thm (Wick) $B_i \in C$, $i=1, \dots, k$.

$$aa^\dagger a^\dagger$$

$$a(x) = A(\delta_x)$$

$$\langle 0 | B_1 B_2 \dots B_k | 0 \rangle = \sum_{\substack{\text{pairings} \\ \text{of } \begin{cases} \text{even} & \text{if } k: \text{odd} \end{cases}}} \prod_{i=1}^{k/2} \langle 0 | B_{\ell_i} B_{\ell'_i} | 0 \rangle.$$

$$\begin{aligned} \langle 0 | B_1 B_2 B_3 B_4 | 0 \rangle &= \langle 0 | B_1 B_2 | 0 \rangle \langle 0 | B_3 B_4 | 0 \rangle \\ &\quad + \langle 0 | B_1 B_3 | 0 \rangle \langle 0 | B_2 B_4 | 0 \rangle \\ &\quad + \langle 0 | B_1 B_4 | 0 \rangle \langle 0 | B_2 B_3 | 0 \rangle. \end{aligned}$$

두개의 공을 계산.

Lemma $B_1 := A(\xi_1) + A^\dagger(\eta_1)$

$$B_2 := A(\xi_2) + A^\dagger(\eta_2)$$

$$\xi_i, \eta_i \in \mathcal{H}$$

$$= L^2(X_m, d\mu_m)$$

$$\langle 0 | B_1 B_2 | 0 \rangle = (\xi_1, \eta_2)$$

pf • Observe $\begin{cases} A(\xi) | 0 \rangle = 0 \\ \text{annihilating empty sp} \\ \langle 0 | A^\dagger(\eta) = 0. \end{cases}$

$$A(\xi) = \sum \alpha_k^* a_k$$

$$\bullet [A(\xi), A^\dagger(\eta)] = (\xi, \eta) 1$$

$$\begin{aligned} \langle 0 | B_1 B_2 | 0 \rangle &= \langle 0 | A(\xi_1) A^\dagger(\eta_2) + A(\xi_1) A(\xi_2) \\ &\quad + A^\dagger(\eta_1) A(\xi_2) + A^\dagger(\eta_1) A(\eta_2) | 0 \rangle \\ &= \langle 0 | A(\xi_1) A^\dagger(\eta_2) | 0 \rangle \end{aligned}$$

$$\begin{aligned} A A^\dagger &= A^\dagger A + [A, A^\dagger] \rightarrow \text{Identity} \\ &= \langle 0 | A^\dagger(\eta_2) A(\xi_1) | 0 \rangle + \langle 0 | [A(\xi_1) A^\dagger(\eta_2)] | 0 \rangle = (\xi_1, \eta_2). \end{aligned}$$

< Proof of Wick's Thm >

$$A|0\rangle = 0$$

$$\langle 0|A^\dagger = 0$$

Induction on k. $B_i = A(\xi_i) + A^\dagger(\eta_i)$

$$\langle 0|B_1 B_2 \dots B_k|0\rangle = \langle 0|B_1 \dots B_{k-1}(A(\xi_k) + A^\dagger(\eta_k))|0\rangle$$

$$= \langle 0|B_1 \dots B_{k-2}(A(\xi_{k-1}) + A^\dagger(\eta_{k-1}))A^\dagger(\eta_k)|0\rangle$$

$$= \langle 0|B_1 \dots B_{k-2}A^\dagger(\eta_k)A(\xi_{k-1})|0\rangle +$$

$$[A(\xi), A^\dagger]$$

$$= \langle 0|B_1 \dots B_{k-2}(\xi_{k-1}, \eta_k)|0\rangle$$

$$[A^\dagger, A^\dagger] = 0$$

annihilating operator commutes.

$$+ \langle 0|B_1 \dots B_{k-2}A^\dagger(\eta_{k-1})A^\dagger(\eta_k)|0\rangle \cdot 0$$

$$(A(\xi_{k-1}) + A^\dagger(\eta_{k-1})) = (\xi_{k-1}, \eta_k) \langle 0|B_1 \dots B_{k-2}|0\rangle$$

$$p = (\omega_p, \vec{p})$$

$$a(p)$$

$$a(\vec{p}) = \frac{a(p)}{\sqrt{2\omega_p}}$$

LEM $\langle 0|a(\vec{p})\phi(x)|0\rangle = \frac{1}{\sqrt{2\omega_p}} e^{i(x,p)}$

$$\langle 0|\phi(x)a^\dagger(\vec{p})|0\rangle = \frac{1}{\sqrt{2\omega_p}} e^{-i(x,p)}$$

$$\langle p|\phi(x) = \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3 2\omega_p} (e^{-i(x,p)} a(p) + e^{i(x,p)} a^\dagger(p))$$

$$= \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3 \sqrt{2\omega_p}} (e^{-i(x,p)} a(\vec{p}) + e^{i(x,p)} a^\dagger(\vec{p}))$$

$d\eta_m$

$$= \int_{\underline{X}_m} e^{-i(x,p)} a(\vec{p}) + e^{i(x,p)} a^\dagger(\vec{p}) d\eta_m$$

mass shell

$$= A(f_x) + A^\dagger(f_x)$$

where $f_x = e^{i(x,p)}$. $A(f) = \int f^* a$.

$$\textcircled{1} \langle 0 | a(\vec{p}) \phi(x) | 0 \rangle$$

$$= \langle 0 | \frac{A(\vec{p})}{\sqrt{2\omega_p}} (A(\vec{p}_x) + A^\dagger(\vec{p}_x)) | 0 \rangle$$

$$= \frac{1}{\sqrt{2\omega_p}} \langle 0 | A(\vec{p}) A^\dagger(\vec{p}_x) | 0 \rangle$$

$$= \frac{1}{\sqrt{2\omega_p}} (\delta_{\vec{p}, \vec{p}_x}) = \frac{1}{\sqrt{2\omega_p}} e^{i\omega_p x}.$$

21. 10.22.

Topic: Calculation of $\langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_1, \vec{p}_2 \rangle$

$$S = 1 + (-ig) \int_{\mathbb{R}^{1,3}} \mathcal{T}(x) dx + \frac{(-ig)^2}{2!} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \mathcal{T}(x_1) \mathcal{T}(x_2) dx_1 dx_2 + \mathcal{O}(g^3)$$

$$\mathcal{T} = \frac{1}{4!} : \varphi(x)^4 :$$

$$\varphi(x) = \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3 \sqrt{2\omega_{\vec{p}}}} (e^{-i(x \cdot p)} a(\vec{p}) + e^{i(x \cdot p)} a^\dagger(\vec{p}))$$

p_1, p_2, p_3, p_4 : distinct.

$$\langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_1, \vec{p}_2 \rangle$$

$$= \langle \vec{p}_3, \vec{p}_4 | \vec{p}_1, \vec{p}_2 \rangle + \frac{(-ig)}{4!} \int_{\mathbb{R}^{1,3}} \langle 0 | a(\vec{p}_3) a(\vec{p}_4) : \varphi(x)^4 : a^\dagger(\vec{p}_1) a^\dagger(\vec{p}_2) | 0 \rangle dx + \frac{(-ig)^2}{2!4!4!} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \langle 0 | a(\vec{p}_3) a(\vec{p}_4) \mathcal{T} : \varphi(x_1)^4 : \varphi(x_2)^4 : a^\dagger(\vec{p}_2) a^\dagger(\vec{p}_1) | 0 \rangle dx_1 dx_2 + \mathcal{O}(g^3)$$

(Wick's Thm)

$$B_i = A(\xi_i) + A^\dagger(\eta_i)$$

$$\langle 0 | B_1 B_2 \cdots B_k | 0 \rangle = \begin{cases} \sum_{\text{pairings}} \pi \langle 0 | B_{l_i} B_{l'_i} | 0 \rangle \\ k: \text{odd.} \end{cases}$$

$$\textcircled{1} \langle 0 | a(\vec{p}_2) a(\vec{p}_4) a^\dagger(\vec{p}_1) a^\dagger(\vec{p}_2) | 0 \rangle = 0$$

$$\langle 0 | a^\dagger = 0 = a | 0 \rangle \rightarrow = A(\delta_{p_q}) \cdot \frac{1}{\sqrt{2\omega_{p_q}}}$$

$$\langle 0 | a(p) a^\dagger(q) | 0 \rangle = \langle 0 | A(\delta_p) A^\dagger(\delta_q) | 0 \rangle = \delta(p-q)$$

$\therefore$ If p_1, p_2, p_3, p_4 are distinct,

$$\langle \vec{p}_3, \vec{p}_4 | \vec{p}_1, \vec{p}_2 \rangle = 0.$$

I. Computation Involving Normal Ordering.

$$\text{Thm } \langle 0 | B_1 \cdots B_i^k \cdots B_m | 0 \rangle$$

$$= \sum \prod \langle 0 | B_l B_k | 0 \rangle$$

pairings
without ones which
has $\langle 0 | B_i B_i | 0 \rangle$.

$$\text{ex) } \langle 0 | B_1 : B_2^2 : B_3 | 0 \rangle$$

$$\langle 0 | B_1 : B_2^2 B_3 | 0 \rangle = \langle 0 | B_1 B_2 | 0 \rangle \langle 0 | B_2 B_3 | 0 \rangle \\ + \langle 0 | B_1 B_2 | 0 \rangle \langle 0 | B_2 B_3 | 0 \rangle \\ - \langle 0 | B_1 B_3 | 0 \rangle \langle 0 | B_2 B_2 | 0 \rangle$$

<Sketch of RP>

$$B_i = A(\xi_i) + A^\dagger(\eta_i), \quad : B_i^k : = \sum_{j=0}^k \binom{k}{j} (A^\dagger)^j A^{k-j}$$

$$\langle 0 | B_1 \cdots : B_i^k : \cdots B_m | 0 \rangle$$

$$= \sum_{j=0}^k \binom{k}{j} \langle 0 | B_1 \cdots (A^\dagger)^j A^{k-j} \cdots B_m | 0 \rangle$$

$$\text{Use } \langle B_1 B_i \rangle \langle B_2 B_i \rangle \langle B_i B_2 \rangle$$

$$\underbrace{\langle B_1 B_i \rangle}_{\#} \\ \langle B A^\dagger \rangle$$

$$\underbrace{\langle B_i B_2 \rangle}_{\#} \\ \langle A_i B_3 \rangle$$

$$\langle \vec{p}_3, \vec{p}_4 : \varphi(x)^4 : \vec{p}_1, \vec{p}_2 \rangle$$

$$= \langle 0 | a(\vec{p}_3) a(\vec{p}_4) : \varphi(x)^4 : a^\dagger(\vec{p}_1) a^\dagger(\vec{p}_2) | 0 \rangle$$

$$\left(\varphi(x) = A(\vec{p}_x) + A^\dagger(\vec{p}_x), \quad f_x(p) = e^{i(x \cdot p)} \text{ Minkowski Inner Product} \right)$$

$$= 4! \langle 0 | a(\vec{p}_3) \varphi(x) | 0 \rangle \langle a(\vec{p}_4) \varphi(x) \rangle$$

$$\langle \varphi(x) a(\vec{p}_1) \rangle \langle \varphi(x) a(\vec{p}_2) \rangle$$

$$\text{Use } \langle 0 | a(\vec{p}) \varphi(x) | 0 \rangle = \frac{1}{\sqrt{2\omega_{\vec{p}}}} e^{i(x \cdot p)}$$

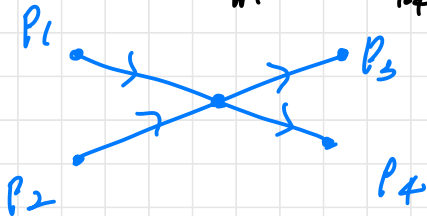
$$\langle 0 | \varphi(x) a^\dagger(\vec{p}) | 0 \rangle = \frac{1}{\sqrt{2\omega_{\vec{p}}}} e^{-i(x \cdot p)}$$

$$= 4! \frac{e^{i(x \cdot p_3 + p_4 - p_1 - p_2)}}{\sqrt{16 \omega_{\vec{p}_1} \omega_{\vec{p}_2} \omega_{\vec{p}_3} \omega_{\vec{p}_4}}}$$

$$\frac{-i g}{4!} \int_{\mathbb{R}^{1,3}} \frac{4! e^{i(x \cdot \dots)}}{\sqrt{16 \dots}} dx$$

$$= \frac{(-i g) \delta(p_3 + p_4 - p_1 - p_2)}{\sqrt{16 \omega_{\vec{p}_1} \dots \omega_{\vec{p}_4}}} \cdot (2\pi)^4$$

$$p_3 + p_4 = p_1 + p_2$$



$$\begin{aligned}
& \langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_1, \vec{p}_2 \rangle \\
&= \langle \vec{p}_3, \vec{p}_4 | \vec{p}_1, \vec{p}_2 \rangle + \frac{(-ig)}{4!} \int_{\mathbb{R}^3} \langle \vec{p}_3, \vec{p}_4 : \phi(x)^4 : \vec{p}_1, \vec{p}_2 \rangle dx \\
&+ \frac{(-ig)^2}{2!4!4!} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \langle \vec{p}_3, \vec{p}_4 : \phi(x_1)^4 : \phi(x_2)^4 : \vec{p}_1, \vec{p}_2 \rangle dx_1 dx_2 \\
&+ \mathcal{O}(g^3) = \mathcal{I}(\text{Type I}) + \mathcal{I}(\text{Type II})
\end{aligned}$$

$$T_A(t) T_B(s) = \begin{cases} T_A(t) T_B(s), & t \geq s \\ T_B(s) T_A(t), & t \leq s. \end{cases}$$

$$T \left(\underbrace{T(t, \vec{x})}_x \right) T \left(\underbrace{T(s, \vec{y})}_y \right) = \begin{cases} T(x) T(y), & t \geq s \\ T(y) T(x), & t \leq s \end{cases}$$

Type I

$$\langle a \phi \rangle \langle a \phi \rangle \langle \tilde{\phi} a^\dagger \rangle \langle \tilde{\phi} a^\dagger \rangle \langle \mathcal{P} \phi \tilde{\phi} \rangle^2$$

Type II

$$\langle a \phi \rangle \langle a^\dagger \phi \rangle \langle \tilde{\phi} a \rangle \langle \tilde{\phi} a^\dagger \rangle \langle \mathcal{P} \phi \tilde{\phi} \rangle^2$$

of Type I choice of ϕ & $\tilde{\phi}$: 2

$$\begin{aligned}
\phi_1, \phi_2, \phi_3, \phi_4 & \leftarrow \text{choice of } \langle a \phi \rangle : 4 \times 3 \\
a, a^\dagger & \text{choice of } \langle \tilde{\phi} a^\dagger \rangle : 4 \times 3 \\
4 \times 3 \text{ (or } 4 \times 3) & \text{choice of } \langle \phi \tilde{\phi} \rangle : 2
\end{aligned}$$

$$2 \times (4 \times 3) \times (4 \times 3) \times 2$$

As a prototype, we compute (type I)

$$\langle 0 | \varphi(x_1) a^\dagger(\vec{p}_1) | 0 \rangle \langle 0 | \varphi(x_1) a^\dagger(\vec{p}_2) | 0 \rangle$$

$$\cdot \langle 0 | a(\vec{p}_3) \varphi(x_2) | 0 \rangle \langle 0 | a(\vec{p}_4) \varphi(x_2) | 0 \rangle$$

$$\cdot \langle 0 | T \varphi(x_1) \varphi(x_2) | 0 \rangle^2$$

But we can $\langle 0 | \varphi(x) a^\dagger(p) | 0 \rangle = \frac{e^{-i(x,p)}}{\sqrt{2\omega_p}}$.

$$\langle 0 | T \varphi(x_1) \varphi(x_2) | 0 \rangle \dots (*)$$

$x_1 = (t_1, \vec{x}_1)$
 $x_2 = (t_2, \vec{x}_2)$

If $t_1 \geq t_2$,

$$\begin{aligned} (*) &= \langle 0 | \varphi(x_1) \varphi(x_2) | 0 \rangle \quad \text{survive} \\ &= \langle 0 | (A(\vec{x}_1) + A^\dagger(\vec{x}_1)) (A(\vec{x}_2) + A^\dagger(\vec{x}_2)) | 0 \rangle \\ &= \langle f_{x_1}, f_{x_2} \rangle \text{ in } L^2(X_m, d\lambda_m) \end{aligned}$$

$$\begin{aligned} \langle f_{x_1}, f_{x_2} \rangle &= \int_{X_m} e^{-i(x_1, p)} e^{i(x_2, p)} d\lambda_m(p) \\ &= \int_{\mathbb{R}^3} \frac{e^{-i(t_1 - t_2)\omega_{\vec{p}} + i(\vec{x}_1 - \vec{x}_2) \cdot \vec{p}}}{(2\pi)^3 2\omega_{\vec{p}}} d\vec{p}. \end{aligned}$$

$$\therefore (*) = \int_{\mathbb{R}^3} \frac{e^{-i|t_1 - t_2|\omega_{\vec{p}} + i(\vec{x}_1 - \vec{x}_2) \cdot \vec{p}}}{(2\pi)^3 2\omega_{\vec{p}}} d\vec{p}.$$

$$=: -i \Delta_F(x_1 - x_2),$$

where $\Delta_F(x)$ is Feynman propagator

$$\Delta_F(x) = i \int_{\mathbb{R}^3} \frac{1}{(2\pi)^3 2\omega_{\vec{p}}} e^{-i\epsilon(\omega_{\vec{p}} + i\vec{x} \cdot \vec{p})} d\vec{p}$$

But $\Delta_F(0) = \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3 2\omega_{\vec{p}}} = \int \frac{d\vec{p}}{\|\vec{p}\|^2 + m^2}$

$\Delta_F(x)$ is a tempered distribution.

$$\int \Delta_F(x) f(x) dx = \int \Delta_F(x) f(x) d\epsilon d\vec{x}$$

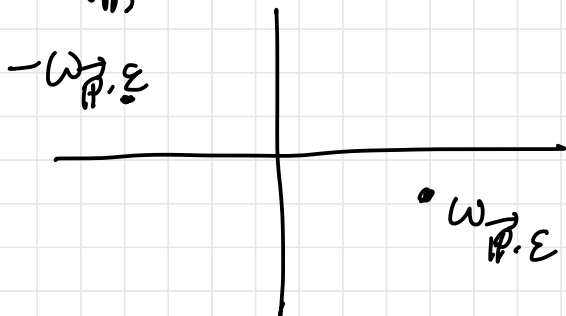
Lemma (Formally) ↗ as a tempered distribution

$$\Delta_F(x) = \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3} \cdot \frac{e^{-i(x \cdot p)}}{-\|p\|^2 + m^2 - i\epsilon}$$

(p.p.)

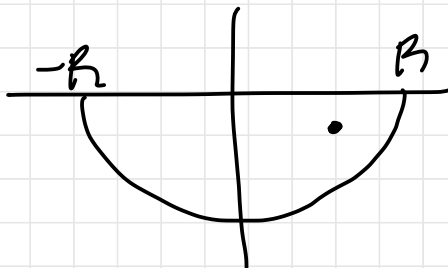
$$\text{RHS} \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^3} \frac{e^{i(\vec{x} \cdot \vec{p})}}{(2\pi)^3} \left(\int_{\mathbb{R}} \frac{e^{-i\epsilon p^0}}{(2\pi) (- (p^0)^2 + \|\vec{p}\|^2 + m^2 - i\epsilon)} dp^0 \right) d\vec{p}$$

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-i\epsilon z}}{f(z)} dz, \quad f(z) = -z^2 + \omega_{\vec{p}}^2 - i\epsilon.$$

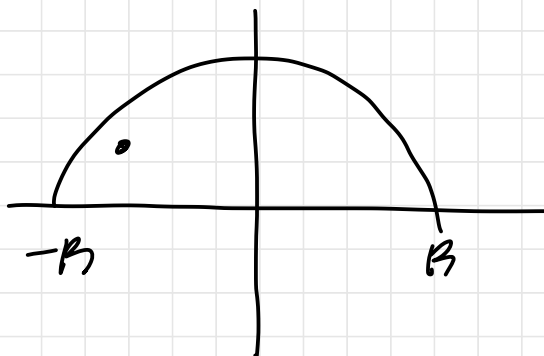


Residue Thm
Jordan Lem.

$t \geq 0$



$t \leq 0$



$$\frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{-i\tau z}}{f(z)} dz = \frac{e^{-i\tau(\omega_{\vec{p}, \varepsilon})}}{2\omega_{\vec{p}, \varepsilon}}$$

inner product
in space
(not Minkowski)

$$\text{R.H.S} = \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^3} \frac{e^{-i\tau(\omega_{\vec{p}, \varepsilon}) + i\vec{x} \cdot \vec{p}}}{(2\pi)^3 2\omega_{\vec{p}, \varepsilon}} = \Delta_F(x).$$

$$\begin{aligned} \langle 0 | \mathcal{T} \varphi(x_1) \varphi(x_2) | 0 \rangle^2 &= \left(-i \Delta_F(x_1 - x_2) \right)^2 \\ &= -\lim_{\varepsilon \rightarrow 0+} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \frac{e^{-i(x_1 - x_2, p)} e^{-i(x_1 - x_2, q)}}{(2\pi)^8 (-\underbrace{\|p\|^2 + m^2}_{\text{Minkowski norm}} - i\varepsilon) (-\|q\|^2 + m^2 - i\varepsilon)} dp dq \end{aligned}$$

$$\langle 0 | \varphi(x_1) \hat{a}^\dagger(\vec{p}_1) | 0 \rangle \langle 0 | \varphi(x_1) \hat{a}^\dagger(\vec{p}_2) | 0 \rangle$$

$$\cdot \langle 0 | \hat{a}(\vec{p}_3) \varphi(x_2) | 0 \rangle \langle 0 | \hat{a}(\vec{p}_4) \varphi(x_2) | 0 \rangle$$

$$\cdot \langle 0 | \mathcal{T} \varphi(x_1) \varphi(x_2) | 0 \rangle = (**).$$

$$(**) = \frac{-1}{\sqrt{16\omega_{\vec{p}_1} \dots \omega_{\vec{p}_4}}} \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}}$$

$$\frac{e^{-i(x_1, p_1 + p_2 + p + q)} e^{-i(x_2, p_3 + p_4 + p + q)}}{(2\pi)^6 (-\|p\|^2 + m^2 - i\epsilon) (-\|q\|^2 + m^2 - i\epsilon)} d^4p d^4q$$

$$\int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \frac{\delta^{(4)}(p_1 + p_2 + p + q) \delta^{(4)}(p_3 + p_4 + p + q)}{(-\|p\|^2 + m^2 - i\epsilon) (-\|q\|^2 + m^2 - i\epsilon)} d^4p d^4q \dots (**)$$

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ix \cdot \xi} d\xi = \delta(x) \quad (\text{Fourier})$$

$$(**) = \int \frac{\delta^{(4)}(p_2 + p_4 - p_1 - p_2) \delta(p)}{(-\|p\|^2 + m^2 - i\epsilon) (-\| -p_1 - p_2 - p \|^2 + m^2 - i\epsilon)}$$

$$\int_{-\infty}^{\infty} \delta(x - z) \delta(y - z) dz = \delta(x - y) \delta(x)$$

In summary,

$$\langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_1, \vec{p}_2 \rangle = (g - i g^2 M + O(g^3)) - \frac{g (2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2)}{\sqrt{16 \omega_{\vec{p}_1} \dots \omega_{\vec{p}_4}}}$$

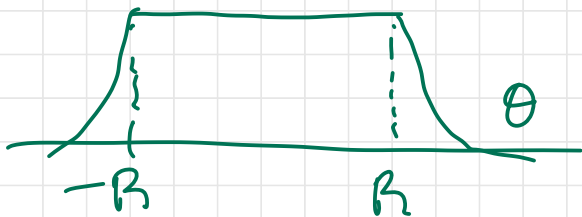
where

$$M = C \int \lim_{\epsilon \rightarrow 0+} \frac{1}{(2\pi)^4} \int_{\mathbb{R}^4} \frac{\theta(\vec{p}) d\vec{p}}{(-\|\vec{p}\|^2 + m^2 - i\epsilon)(-\|\vec{p}_1 + \vec{p}_2 - \vec{p}\|^2 + m^2 - i\epsilon)}$$

$\sim \int \frac{1}{\|\vec{p}\|^4} = \text{div}$
 NO WAY!
 change of variable
 $p \rightarrow -p$

So we add

where $\theta(\vec{p}) = 1$ if $|\vec{p}| < R$ and decay fast outside of it.



Renormalization

Problem : We don't know θ .

Sol : We don't have to know, provided
that for special $(p_1^*, p_2^*, p_3^*, p_4^*)$
if we know $g - i\gamma \tilde{M}_* + \mathcal{O}(g^3)$.